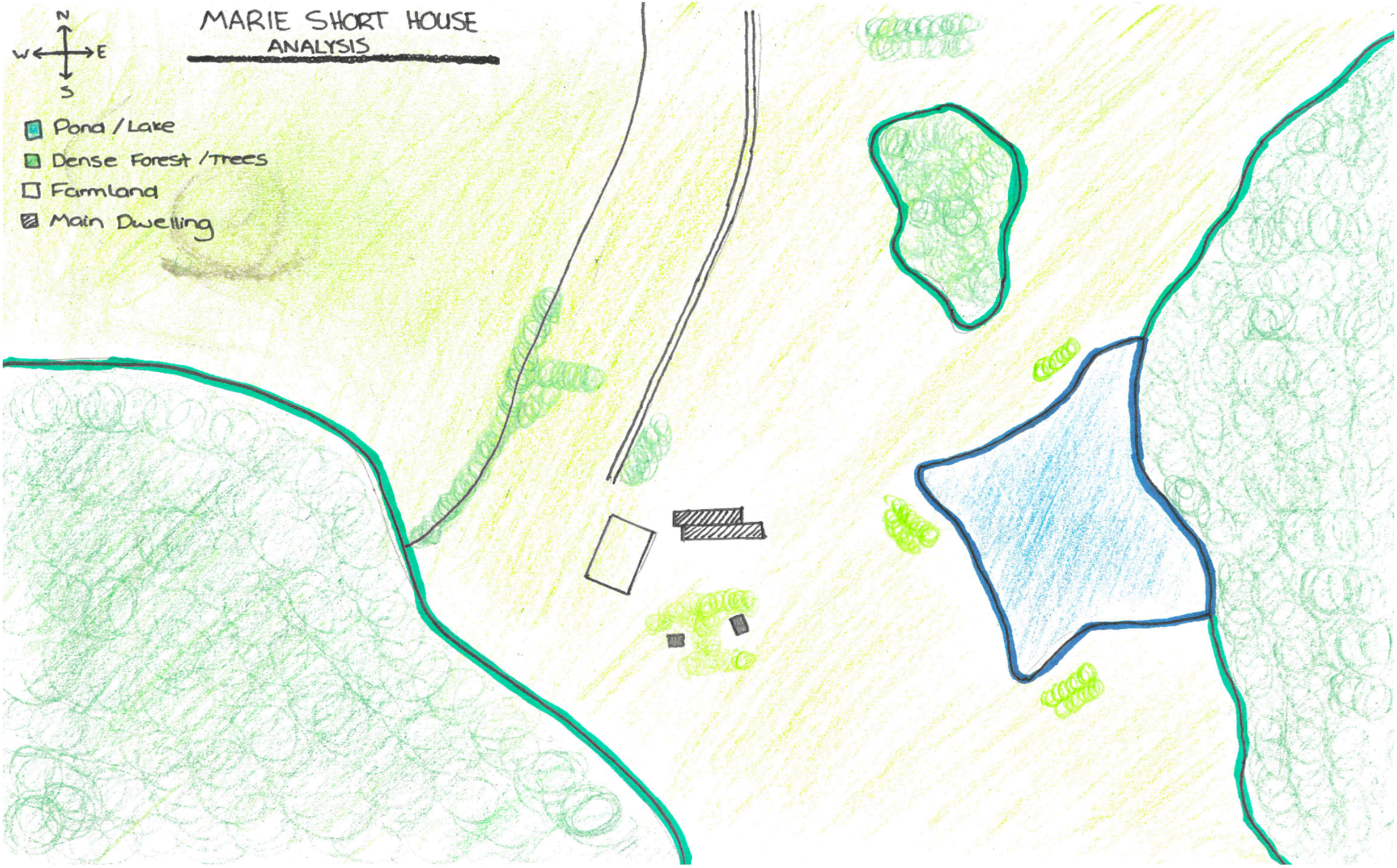
Portfolio A and B – Marie Short
House designed by 'Glenn Murcutt'

Presented by Fergus Shaw

Analysis Diagrams

- A draft analysis drawing I completed prior to beginning on photoshop.



MARIE SHORT HOUSE DESIGNED BY GLENN MURCUTT

The location of the Marie Short house is situated on a rural farm land. The pond located on the eastern side of the property allows for cooler winds to be blown in, therefore during summer months cools the house. The large foresty trees to the southern side reduce the amounts of wind blown in during winter months, allowing the house to stay warmer.

The louvres seen in the photograph above allow for cross venitlation from winds coming north - south.

Property and planning overlays:
- RU1 (Primary Production)
- RU2 (Rural Landscape)
- BMO (Bushfire Prone Land)
- HO (Heritage Overlay)
- VPO (Vegetation Prone Overlay)

Site Details:
- Latitude: 31°04S,
- Longitude: 152°50E
- Climate Zone: Humid subtropical climate
- Total Area: 3,308km²
- Elevation: approx 10m above sea level

Vegetation:
The foresty vegetation surrounding the property allow for natural habitats for local wild life.

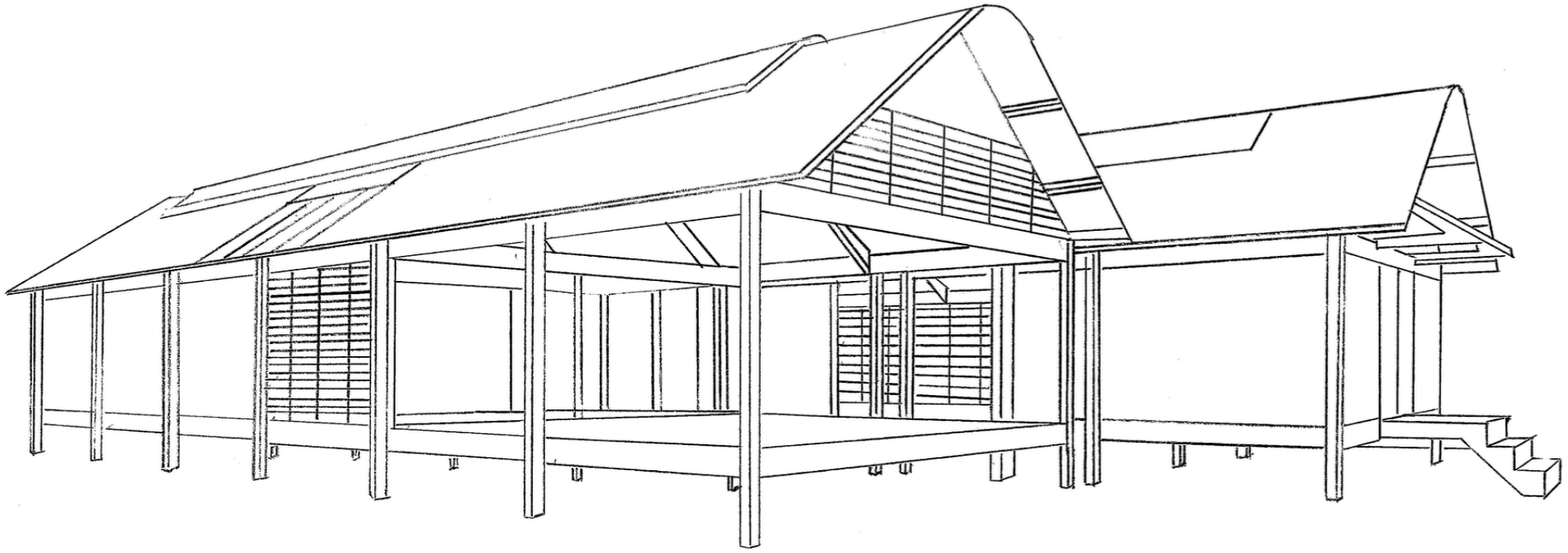
Also, the bushfire overlay indicated above, suggests due to the large amounts of trees that it may be a BAL 19 to BAL 40. Therefore, the Murcutt would have needed to design the house to these measures.

SUN PATH DETAILS:
Murcutt designed the house to a specific orientation. The living pavillion faces north to recieve sunlight for majority of the day. North facing windows haver been specifically placed to attract the warm sunlight. Furthermore, has created open farm land to the north to ensure no sunlight is blocked from large trees.

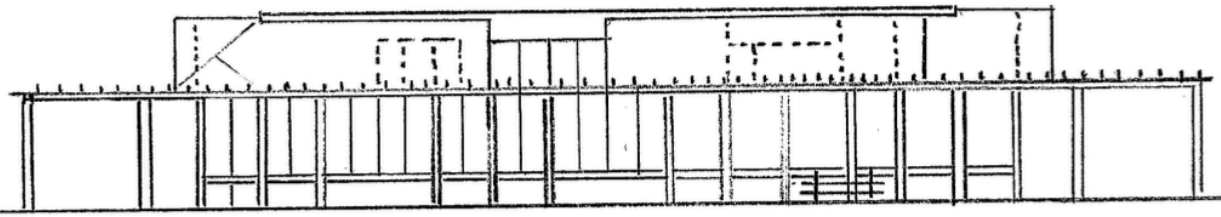
Perspective Drawings

PERSPECTIVE 1 : Marie Short House

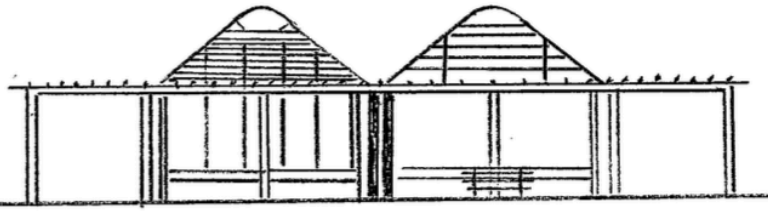
by Glenn Murcutt



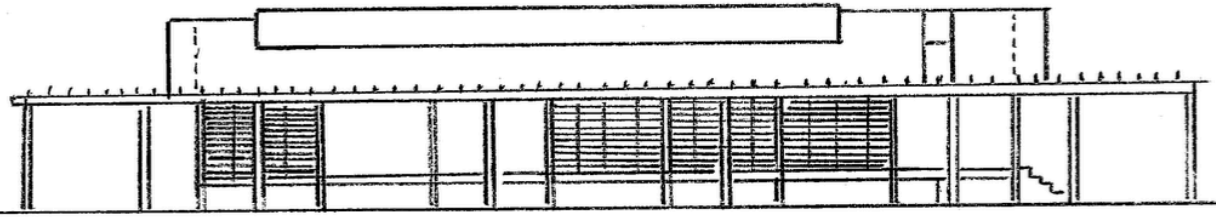
ELEVATION DRAWINGS:



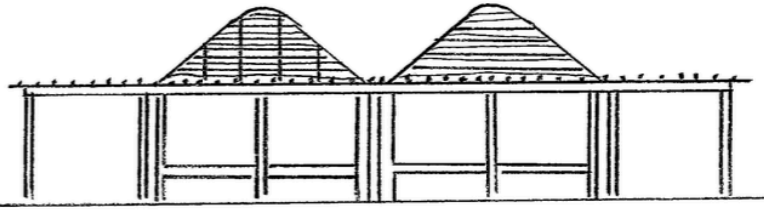
North



East



South

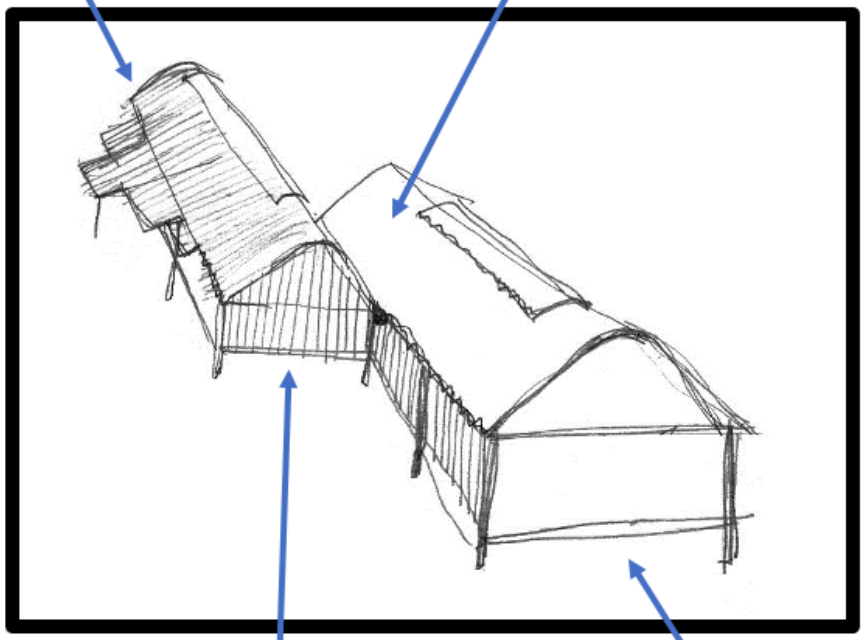


West

Model Making Plan – Marie Short House

Plywood will be used for all external walls

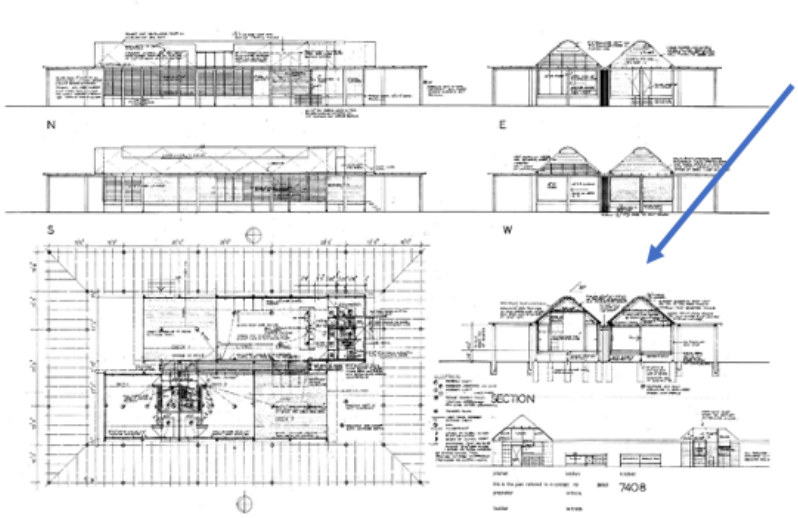
The roofing material will need to be bent in order to achieve an accurate representation. A lightweight tin or aluminum could be used to achieve this. The roofing will be separate from the walls and frame. This will allow it to be taken off/removed.



Timber louvre system – I will used left of plywood and cut them to size and stick them with super glue.

Plywood flooring to be purchased from bunnings. Approx 4mm thick.

The fireplace flu will be achieved by a steel tube from Bunnings.



The plan set the model will be based on. Can be found here: https://www.atlasofplaces.com/architecture/marie-short-house/

Budget for this model – should be around $50. I already own the power tools. So all I need is the materials from Bunnings. Materials include:

- Plywood
- Glue/super glue
- Steel tube
- Tin / Aluminum sheeting
- Level

THEME

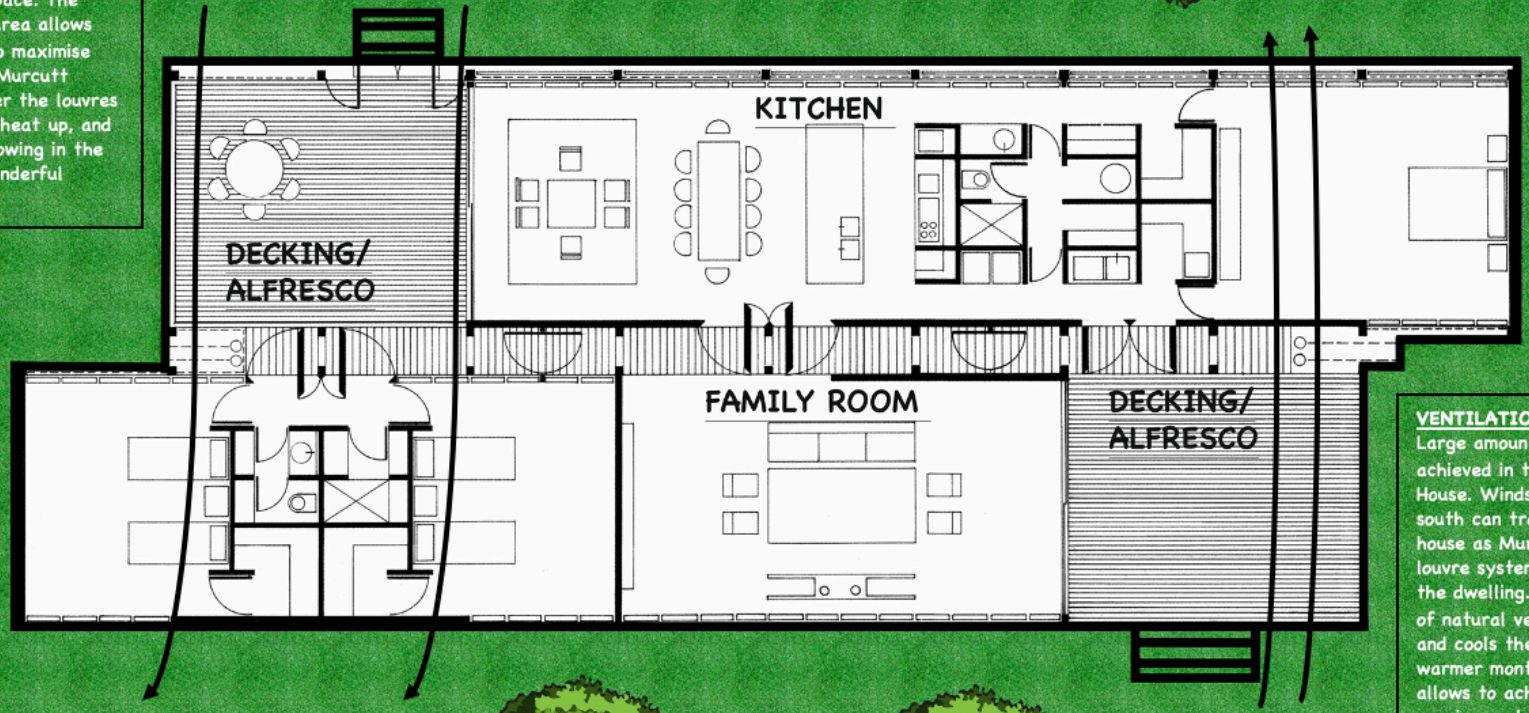
Our group has decided to base our case studies of the famous Australian architect Glenn Murcutt. Majority of his designs/houses are situated in rural farmland, forest like areas, so we have decided to choose from these houses. The photo shop theme we have agreed upon is having a black background (minimalistic) with the site photograph in the middle. We will continue upcoming portfolios and presentations with a similar style i.e the black simple back ground.

Also we have chosen to use the Chalkboard font. Furthermore, have decided to focus on Murcutts building materials for our final theme.

PORTFOLIO PART B – FLOOR PLAN ANALYSIS

MARIE SHORT HOUSE – FLOOR PLAN

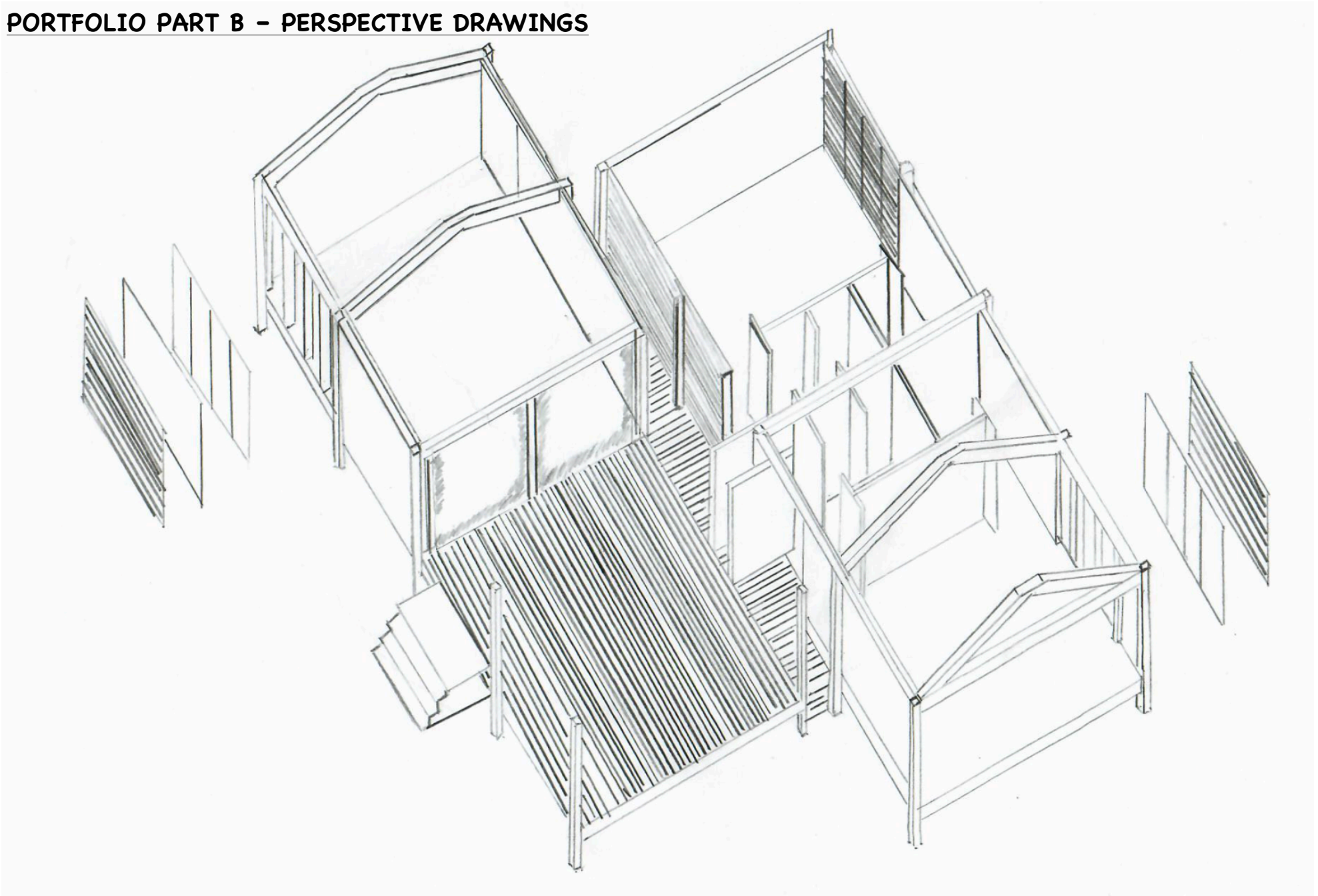
Murcutts Design: Glenn Murcutt designed the house around the concept of maximising the sunlight from the north. This has been achieved by large windows and a louvre system to the north kitchen and living space. The decking / alfresco area allows for a great space to maximise sunlight and relax. Murcutt quotes "in the winter the louvres have a tendency to heat up, and when the wind is blowing in the summer, it has a wonderful cooling effect".



VENTILATION (black arrows): Large amounts of ventilation is achieved in the Marie Short House. Winds from the north and south can travel through the house as Murcutt has installed a louvre system on both sides of the dwelling. This allows plenty of natural ventilation to occur and cools the house down during warmer months. Furthermore, allows to achieve the NCC requirement of a minimum of 5% ventilation.

N

PORTFOLIO PART B – PERSPECTIVE DRAWINGS



PHOTOSHOP - REGIONAL ANALYSIS

MARIE SHORT HOUSE - REGIONAL ANALYSIS

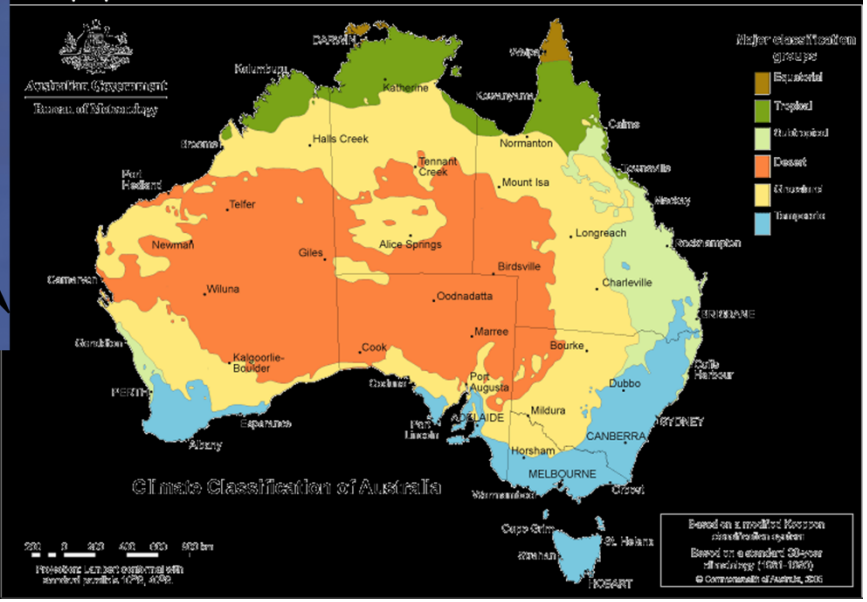
NATIONAL PARK

CRESCENT HEAD OCEAN

SOUTH KEMPSEY APPROX 22 KM NORTH WEST.

A1 - PACIFIC HIGHWAY

CLIMATE ZONE: As per the graph below Kempsey and Crescent head fall into a Sub-Tropical climate zone. Located approx a 6 hour north drive from Coffs Harbour, the Marie Short house lies. Due to the geographical location, the property experiences extreme rainfall during the wet seasons, hence Murcutt has designed for this, by building the house on stumps/posts.



TOPOGRAPHY: Located 22km to the north the town Kempsey consists of the Maclaey River running through the centre of it out into the ocean. This river also runs through a section of the western side of the Marie Short Property. The Maria National Park is located to the north of the MS house and provides shelter to a variety of flora and fauna.

The property sits on pretty flat ground above a marshy sub-tropic flood plain, and is surrounded by an abundance of large trees and grass lands.

Model Making Progress: I have purchased all the material for the model and cut all pieces to size. I just now to assemble the model together. I am using a hot glue gun to attach pieces and a Stanley knife / circular saw to cut other larger pieces. Both floors of the Marie Short house have been complete, and I am working on the posts at the moment.